Joshua N. Meeks

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Education:

University of Nevada, Reno — Graduation: December 2026
B.S. Mechanical Engineering

Engineering Experience:

Digital Solid State Propulsion [*Engineering Intern*] Reno, Nevada — June 2025 – Present

- Designed fuel feed system for F-16 emergency power unit gas generator test bed.
- Designed and fabricated a power and signal distribution system for test stand valves, and DAQ systems.
- Supported development of ~500V electrical solid propellant control boards and charging systems.
- Supported development of hydroxylammonium nitrate (HAN) based propellant systems.
- Designed and fabricated custom test equipment, including a positive pressure glovebox for catalyst testing and an ABL friction tester and vacuum thrust stand for propellant characterization.
- Conducted strand burn rate and hot fire testing.
- Developed MIL-STD based experimental procedures.
- Developed compostable training cartridges for military tank simulations.

UNR Aerospace Club [*Vice President*] — September 2025 – Present

- Developed custom flight computer for rocket telemetry and flight data acquisition.
- Established in house PCB fabrication using CNC milling and a custom PID controlled reflow oven.
- Contributed to liquid rocket and thrust stand development, testing, and organization.

Innevation Center [*Student Worker*] | Reno, Nevada — March 2024 – September 2025

- Automated business data analysis for improved accuracy and time savings.
- Co-developed a new summer robotics camp for kids 4-8 years old.
- Taught students to use shop machines such as lathes, laser cutters, and woodshop tools.

Acree Technologies [*Engineering Intern]* | Concord, California — Summers/Breaks 2018 - 2023

- Assembled and programmed PLC system to automate deposition process.
- Implemented Modbus communication between LabVIEW and PLC Ladder Logic for automation control.
- Analyzed experimental coatings with tribometers to refine production processes.
- Supported development and fabrication of new deposition heads that reduced production time by 50%.
- Performed overhauls on vacuum systems using roughing, turbomolecular, and diffusion pumps.

Technical Skills

- CAD: SolidWorks (CSWA), Creo, Onshape, KiCad.
- Analysis: SolidWorks FEA, COMSOL, OpenMotor, ProPep.
- Programming: C++, HTML, MATLAB, LabVIEW, PLC Ladder Logic.
- Instrumentation: NI DAQ systems, high temperature and pressure sensing, ESP32, Arduino.
- Manufacturing: CNC Milling, 3D Printing, Laser Cutting, Arc Welding.